Research on Cybersecurity Domains | Muhammad Ahmad

Objective

To help students understand various cybersecurity domains, career paths, required skills, and industry certifications.

1) Malware Analysis

Definition [1]

Malware analysis is the study of the unique features, objectives, sources, and potential effects of harmful software and code, such as spyware, viruses, malvertising, and ransomware.

By using static and dynamic techniques in secure, isolated environments, security analysts identify indicators of compromise (IOCs) through the structured process of dissecting malicious software, to determine the severity of threats, to understand its functionality, origin, and impact as well as create detection strategies

It analyzes malware code to understand how it varies from other kinds.

Responsibilities [2]

- Indicator of Compromise (IOC) extraction
- Threat hunting and script automating
- Static and dynamic threat analysis and reverse engineering
- Malware triage and sandbox testing
- Collaborating with threat intel and SOC teams
- Generating and Provide technical reports to internal teams

Technical Skills [3][4]

- Strong knowledge of C/C++,Assembly (x86/x64, ARM), Windows API, and OS internals
- Understand unpacking, deobfuscation, and anti-debugging techniques
- Reconstruct Registry, APIs, PE and unknown file formats & data structures
- Python, Perl, Ruby scripting
- Static & Dynamic Analysis in isolated and sandbox environments
- Threat intelligence, network, protocols and memory analysis

Soft Skills [4]

- Pattern Recognition.
- Root Cause Analysis and Logical Deduction
- Technical Writing
- Collaboration and Information Sharing
- Hypothesis Testing

Tools [5]

- RegShot
- IDA Pro
- Comae-ToolkitCuckoo Sandbox\
- Cuckoo Sandbox
- CFF Explorer
- pestudio

Certifications [6][7][8]

- GIAC Reverse Engineering Malware (GREM)
- Blue Team Level 2 (BTL2)
- Practical Malware Research Professional (PMRP)

Career Path [9]

SOC Analyst → Malware Analyst → Malware Engineer

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2) Penetration Testing

Definition [10]

Pen testing is the planned cyberattacks against computer systems and infrastructure to pinpoint vulnerabilities and weak points which are then fixed.

It is an authorized, simulated attack conducted by security experts and the purpose of this simulated attack is to identify, exploit, and remediate vulnerabilities in networks, applications, and systems before malicious hackers do. It helps identifying any weak spots in a system's defenses which attackers could take advantage of.

It acts as a proactive defense mechanism to uncover weaknesses, ensure regulatory compliance, and strengthen overall security posture.

Key Responsibilities [11]

- Perform automatic and manual attacks
- Exploit security weaknesses ethically
- Perform network testing
- Document reports and findings
- Suggest improvements and upgrades

Required Technical Skills [12]

- Networking Fundamentals (IDS, IPS)
- Linux & Windows administration
- Scripting (Python, Bash, PowerShell)

Required Soft Skills [13]

- Analytical thinking
- Teamwork
- Report writing
- Communication with technical & non-technical teams

Tools Used [14]

- Metasploit
- Burp Suite
- Nmap
- Wireshark
- Kali Linux

Certifications [15]

- CEH – by EC-Council
- OSCP – by Offensive Security
- Security+ – by CompTIA
- GPEN – by GIAC

Career Path & Growth [16]

Junior Pentester → Senior Pentester → Red Team Lead → Security Consultant → Security Architect

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3) Digital Forensics

Definition [17]

Digital Forensics involves identifying, preserving, analyzing, and presenting digital evidence related to cybercrime in a way that it maintains its integrity in the court.

Digital forensics is a field of forensic science where digital and electronic data is scientifically identified, preserved, examined, and analyzed to reconstruct incidents, ensuring evidence remains admissible in legal proceedings. It is used to investigate cybercrimes but can also help with criminal

and civil investigations. Cybersecurity teams can use digital forensics to identify the cybercriminals behind a malware attack, while law enforcement agencies might use it to analyze data from the devices of a murder suspect.

It covers computers, mobile devices, and networks to fight cybercrime, insider threats, and data breaches. Key steps include securing evidence and maintaining a chain of custody

Key Responsibilities [17]

- Evidence acquisition & chain of custody
- Disk & memory analysis
- Log examination
- Data Recovery

Technical Skills [18]

- File systems and disk imaging
- Memory forensics
- Network log analysis
- Legal & compliance procedures

Soft Skills [19]

- Investigative and problem solving
- Documentation accuracy
- Legal awareness
- Critical thinking

Tools [18]

- Xplico
- FTK Imager
- Sleuth Kit

Certifications [19]

- Certified Computer Examiner (CCE)
- GIAC Certified Forensic Analyst (GCFA)
- GIAC Certified Forensic Examiner (GCFE)

Career Path [20]

Ethical hacking essential E|HE → Certified Cybersecurity Technician C|CT → Certified Network Defender C|ND → Certified Incident Handler C|CIH

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4) SOC Analyst

Definition [21]

SOC stands for Security Operations Center and a SOC analyst is a person who monitors, analyzes, and responds to security issues.

A SOC Analyst is a force multipler frontline cybersecurity professional responsible for monitoring, detecting, and responding to security threats in real-time. They analyze security alerts from automated systems like SIEM, ITDR and EDR to mitigate risks, investigate incidents, and protect an organization's digital.

With cyberattacks constantly evolving and the variety of customer environments they encounter, SOC analysts have to be adaptable and continuously learn to stay ahead of threat actors' tactics, techniques, and procedures.

Responsibilities [21]

- Surveillance of an Organization's Network
- Investigation and identification of attacks
- Responding efficiently to attacks
- Security audits and attack documentations

Technical Skills [21][23]

- Networking fundamentals
- Splunk, IBM QRadar
- Python and PowerShell scripting

Soft Skills [21]

- Problem solving
- Quick decision-making
- Stress management

Tools [22]

- Real Intelligence Threat Analytics (RITA)
- Suricata
- Snort
- SIEM
- EDRs

Certifications [21]

- CompTIA Security+
- CompTIA CySA+
- Certified SOC Analyst (CSA) by EC-Council

Career Growth [24]

SOC Tier 1 → Tier 2 → Tier 3 → SOC Team Lead → Threat Hunter

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5) Identity and Access Management

Definition [25]

Identity & Access Management is the framework of securing digital identities with the help of authorization and authentication. This need for security extends to vendors, contractors and partners of the organization. It consists of two parts: Identity Management and Access Management.

It enhances security, ensures regulatory compliance, and supports cloud/remote work by managing digital identities, authenticating users, and enforcing authorization.

Responsibilities [25][26]

- Defense against data breaches
- Data encryption
- Password protection
- Behavioral & static analysis
- Manage identity databases

Technical Skills [26]

- Privileged Access Management
- Risk based Authentication
- Data Governance
- Federated Identity Management

Soft Skills [27]

- Communication
- Collaboration
- Customer Service

Tools [25]

- OpenID Connect (OIDC)
- System for Cross-Domain Identity Management (SCIM)
- Security Assertion Markup Language (SAML)

Certifications [27]

- Certified Identity and Access Manager (CIAM)
- Identity and Access Administrator Associate (SC-300)
- Certified Identity Management Professional (CIMP)

Career Path [28]

IAM Analyst → IAM Engineer → IAM Architect

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6) Governance, Risk and Compliance

Definition [29][30]

Governance, Risk, and Compliance (GRC) is a strategic framework to align IT with business governance with risk management while meeting all industry and government regulations. It includes tools and processes to unify an organization's governance and risk management with its technological innovation and adoption.

GRC combines governance, risk management, and compliance in one coordinated model. This helps reduce wastage, increase efficiency, reduce noncompliance risk, and share information more effectively.

Responsibilities [29][30]

- **Governance :**
  - Utilizing risk-related insights
  - Defining conflict resolution policies and procedures
- **Risk management** :
  - Predicting threats and their potential damage
  - Analyzing legal, financial and security risk
- **Compliance** :
  - Aligning internal corporate as well as business policies with regulatory standards

Technical Skills [31]

- ISO 27001 and SOC 2 framework
- NIST cybersecurity framework
- Cyber threat intelligence

Soft Skills [32]

- Documentation
- Risk management
- Stakeholder management

Tools [31]

- ServiceNow GRC
- RSA Archer
- LogicGate Risk Cloud

Certifications [33]

- GRC Professional (GRCP)
- GRC Auditor (GRCA)
- Integrated Audit and Assurance Professional (IAAP)

Career Path [34]

Cybersecurity Auditor → GRC Analyst → GRC Director → GRC Architect

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7) Application Security (AppSec)

Definition [35][36]

Application security are the measures, practices and tools designed to identify software applications threats and vulnerabilities in software applications throughout their SDLC life cycle.

Application security aims to protect software application code and data against cyber threats during all phases of development, including design, development, and deployment.

Responsibilities [36]

- Conducting a Risk Assessment
- Integrating Security into the Development Lifecycle
- Performing Regular Audits and Updates
- Automating Testing Tools
- Planning for Incidents

Technical Skills [37]

- Proficiency in Java, C++, Python, and Ruby as well as other programming languages
- Familiarity with security frameworks and standards such as OWASP Top Ten ISO 27001 information security management system (ISMS)
- Penetration testing and cloud security

Soft Skills [37]

- Teamwork and Communication
- Critical thinking
- Documentation
- Continuous learning

Tools [38]

- Semgrep
- SonarQube
- OWASP Dependency-Check
- SQLmap

Certifications [39]

- Certified Application Security Engineer (Case JAVA)
- Web Application Hacking and Security (WAHS)
- Certified Identity Management Professional (CIMP)

Career Path [40]

AppSec Associate → DevSecOps Engineer → AppSec Architect

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8) Cyber Threat Intelligence

Definition [41]

Threat intelligence is the analysis of data using tools and techniques to generate meaningful information about existing or emerging threats targeting the organization that helps mitigate risks including threat actor motives, targets, and methods.

Threat Intelligence helps organizations make faster, more informed security decisions and change their behavior from reactive to proactive to combat the attacks this analyzed, intelligence based knowledge enables organizations to move from reactive defense to a proactive, informed security posture. It transforms raw data into actionable insights to help security teams identify, prevent, and mitigate risks.

Responsibilities [41][42]

- Identifying organizational threat landscape
- Data Collection through Open Source Intelligence (OSINT) and Human Intelligence (HUMINT)
- Bulk data analysis, research and exploitation
- Intelligence reporting and dissemination

Technical Skills [43][44]

- Familiarity with threat intelligence frameworks like MITRE ATT&CK
- Familiarity with Threat intelligence platforms (TIP) and data collection SIEMs and XDRs
- Malware analysis and reverse engineering
- Shell and Python scripting and automation

Soft Skills [45]

- Analysis skills
- Critical thinking
- Creativity and pattern recognition

Tools [46]

- Recorded Future Intelligence
- MISP Threat Sharing
- YETI
- OpenCTI

Certifications [45][47]

- Information Systems Security Engineering Professional (ISSEP)
- Global Information Assurance Certification (GIAC)
- Joint Cyber Analysis Course (JCAC)
- PECB Certified Cyber Threat Analyst (CCTA)

Career Path [48]

Vulnerability/Threat Analyst → Cyber Threat Intelligence Analyst → Security Researcher, Threat Hunter

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9) Network Security

Definition [49]

Network security is a set of practices and technologies that keep internal networks protected from attacks and data breaches. It focuses on access control, cyber attack prevention, malware detection, and other security measures, protecting computer networks and communication systems from internal and external cyberthreats and cyberattacks.

Responsibilities [49][50]

- Setting up authorization and authentication policies
- Enforcing multifactor authentication (MFA)
- Building a zero-trust framework
- Threat detection and analyzing security risks
- Incident Response and recovery

Technical Skills [51]

- Securing networks through firewall management, IDPS, VPNs, SIEM
- Setting up security protocols, endpoints, PKI, NAC and cloud security solutions.
- Configuring Network Access Control (NAC), Network Protocols and Packet Analysis
- Cyber Threat intelligence and research
- Python, PowerShell, or Bash scripting

Soft Skills [51]

- Critical Thinking and Problem Solving
- Effective Communication
- Team Collaboration
- Adaptability and Flexibility
- Customer Service

Tools [52][53]

- Wireshark
- Metasploit
- Nessus
- Aircrack
- Splunk SIEM
- Nmap

Certifications [54][55]

- Certified Network Defender (CND)
- Certified Ethical Hacking (CEH)
- ComPTIA Security+ (Sec+)
- Cisco Certified Network Associate (CCNA)

Career Path [56]

Security Analyst → SOC Analyst/PenTester → Chief Information Security Offier

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10) Cloud Security

Definition [57]

Cloud security refers to the comprehensive set of cybersecurity policies, best practices, controls, and technologies designed and used to protect applications, data, and infrastructure in cloud environments.

In particular, cloud security works to provide storage and network protection against internal and external threats, access management, data governance and compliance, and disaster recovery. It secures cloud-based services against unauthorized access, data breaches, and cyber threats through encryption, identity management, and threat detection

It specifically uses “shared responsibility" approach, where security is ensured “of the cloud” and of data “in the cloud”

Responsibilities [58][59]

- **”Security of the Cloud”**
    - Cloud providers secure the underlying infrastructure of the cloud
    - Enforcing security measures like least-privilege access or API protection.
    - Securing virtual machines and infrastructure
    - Compliance with the cloud security policies and standards
- **”Security in the Cloud”**
    - Users secure their data and connections with the cloud
    - Endpoint security measures and firewall configurations
    - Identity and access management and application security

Security responsibilities shift dramatically depending on the cloud types IaaS, SaaS or PaaS.

Technical Skills [59][60]

- Cloud platform and server knowledge
- VM, Container and orchestration security knowledge
- Linux, Python, Bash, or PowerShell Scripting and DevOps Automation
- CI/CD & automation
- Incident Response and Forensics
- Identity Management and Access
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Soft Skills [60]

- Communication and presentation
- Team work and Collaboration
- Problem solving and analytics

Tools [61][62]

- Cloud Native Application Protection Platforms (CNAPP)
- SentinelOne Singularity Cloud Security
- Cloud infrastructure entitlement management (CIEM)

Certifications [63][64]

- Certified Cloud Security Professional
- Azure Security Engineer Associate
- AWS Certified Security

Career Path [65]

Cloud Support Technician → Cloud Security Engineer → Cloud Security Architect

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